Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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| **Proposer** | **Name** | Huachao Mao | **Email** | coria-consortium@purdue.edu |
| **Institute Name** | Purdue University | | | |
| **Address** | Knoy 125, 401 N. Grant Street, West Lafayette, IN 47907 | | | |

## 2. Technology Overview

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| **Title of Proposed Technology** | **Application of AI Technologies to Enhance Transportation Efficiency and Durability of Hovercraft and Vessels, and Commercial Development of Ultra-Lightweight Composite PTM and Ship-Specific Containers** | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| **Industrial Technology Classification** [1] | **Technology Field 1 (**Large Category**)** | | **Technology Field 2 (**Large Category**)** | |
| | *Machinery and materials* | | | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Shipbuilding/Marine Systems Software* | *Ship Materials/Structural Technologies* | | |
| **Concept of Proposed Technology** | **Core Concept**<br>The proposed concept focuses on the Personnel Transport Module (PTM) for military hovercraft is a modular system designed to maximize personnel transport capacity for high-speed amphibious landing craft.<br>The project aims to develop and mass-produce composite-material Personnel Transport Modules (PTMs) to replace existing aluminum (AL)-based PTMs, reflecting the operational requirements of military end users. In addition to military applications, the composite PTMs will also be developed for civilian uses, including tourism, rescue operations, and marine transportation, followed by domestic and international promotion and commercialization.<br>Currently, in Korea, Sejin Technology Industry Co., Ltd. manufactures aluminum-based PTMs and ship containers, while the Navy and Marine Corps are increasingly demanding weight reduction and enhanced durability, highlighting the need for next-generation composite PTM solutions. | | | |

**Key Functions and Potential Applications**

- To maximize buoyancy, which is a core performance factor of hovercraft, the proposed technology focuses on converting the Personnel Transport Module (PTM)—which carries onboard personnel—and cargo containers from conventional aluminum/steel structures to sandwich-structured composite materials.
- Development of a PTM to enhance the performance of LCAC platforms, meeting the operational requirements of the Navy
- Achievement of key performance goals, including improved durability, enhanced corrosion resistance, weight reduction, improved sound insulation performance, and reduced life-cycle costs
- Application of the developed technology to general ship containers as well

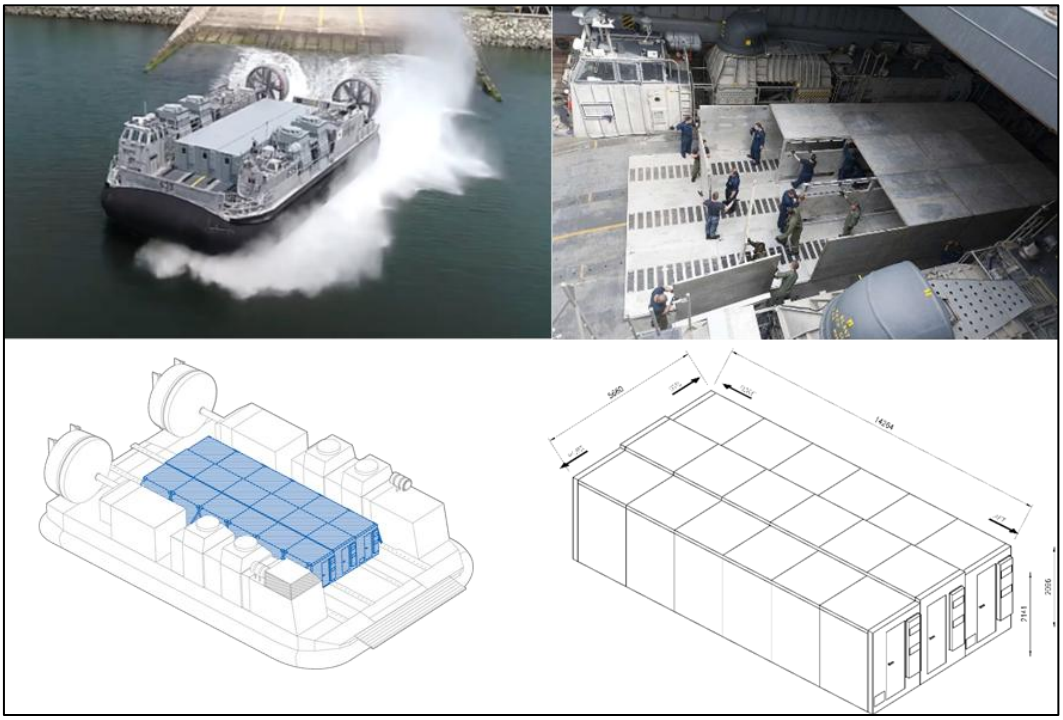

**Figure. PTM for Hovercraft**

**Figure. Currently operating aluminum (AL)-based PTM**

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| **Rationale for Joint R&D Project** | **Critical Technical Barriers**<br>The manufacturers of UHMWPE include Honeywell in the United States (trade name: Spectra), DSM in the Netherlands (trade name: Dyneema), and Chinese suppliers such as DDZM and BJTYZ. Korea currently relies entirely on imports for UHMWPE, primarily importing products from the Netherlands for the production of ballistic materials, highlighting the need for localization and domestic production.<br><br>**Why Global Collaboration?**<br>Collaboration is essential between an overseas partner possessing core technologies for the raw material manufacturing of UHMWPE (Ultra-High-Molecular-Weight Polyethylene), a high-performance ultra-lightweight composite material, as well as fundamental technologies for hybrid composite structural analysis, and a Korean composite manufacturing specialist capable of applying these technologies to hovercraft and vessels and integrating them at the system level.<br><br>**Expected Impact**<br>Through collaboration with the overseas research partner, the project will establish:<br>● Manufacturing AX is critical to corporate survival, making AI-based technology development essential. This project improves technology development efficiency by using AI to analyze patents, publications, and experimental data, and to support virtual experiments and outcome prediction.<br>● AI is applied to operator training and to work, process, and shipment inspections to enhance production efficiency and reduce defects by monitoring manufacturing conditions and workplace environments. These efforts support improvements in productivity, safety, quality assurance, cost reduction, and process automation.<br>● AI-based design, analysis, and simulation—including protective performance and flow analysis—are used to reduce time and cost and to enable AI-driven product reverse engineering.<br>● UHMWPE raw materials are domestically developed for experimental use, followed by fabrication of specimens and prototypes and certification testing to link AI-based technology development with material-level validation. |
| **Key Words** | 1. UHMWPE Fabric & Film<br>2. Hovercraft<br>3. PTM (Personnel Transport Module)<br>4. Composite container<br>5. Structural lightweighting |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 4 ) | End of the Project | Level ( 7 ) |
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| R&D Objectives | **R&D Objectives**<br>1) Structural design and analysis for the development of prototypes of a 50-person-class composite PTM for hovercraft installation and a 20-ft-class ultra-lightweight cargo container<br>2) 3D modeling (hybrid type)<br>3) Specimen fabrication and testing (including tests conducted in the United States)<br>4) Fabrication and evaluation (performance review) of scaled-down prototypes | | | |
| R&D Target Specifications | **Performance Targets and Validation Criteria**<br>1) Weight reduction: Achieving at least 25% weight reduction compared to conventional aluminum structures<br>2) Durability: Ensuring zero corrosion in marine saltwater environments and securing impact resistance<br>3) Safety: Compliance with the International Maritime Organization (IMO) fire safety requirements (FTP Code)<br>4) Environmental test requirements:<br>● Moisture absorption: Applicable standard (MIL-PRF-64154A), test method (ASTM D570)<br>● Flame propagation: Applicable standard (MIL-STD-1623), test method (ASTM E162)<br>● Combustion gases: Applicable standard (MIL-STD-2031), test method (ASTM E1354)<br><br>**Composite material selection and structural design**<br>1) Key materials: High-strength carbon fiber–reinforced polymer (CFRP) for the PTM outer skin, combined with UHMWPE (fabric and film), which provides high impact resistance, excellent protective performance, and effective weight reduction.<br>2) Sandwich core: Application of aluminum honeycomb or high-density PVC foam cores.<br>3) Forming process: Use of VaRTM (Vacuum-assisted Resin Transfer Molding) to minimize void formation and optimize structural strength.<br><br>**Quantitative Targets**<br>1) Target performance required by end users (Korean and U.S. Navy/Marine Corps)<br>2) Target performance required for general vessels, including mechanical and physical properties, weight and corrosion resistance, flame-retardant performance, etc. | | | |

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| **R&D Plan** | **Year 1: Design of composite sandwich panels and fire/impact simulations**<br><br>**Year 2: Development of PTM modular assembly technology and container fastening systems**<br><br>**Year 3: Noise, vibration, and stability validation through installation on an actual hovercraft, and acquisition of relevant certifications** |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | 1) Support for optimal formulation and lay-up design of high-performance composite materials (GFRP/CFRP)<br>2) Structural strength analysis and data analysis, including measurement of elastic modulus<br>3) Analysis of impact energy absorption of composite panels and fatigue life prediction modeling<br>4) Technical support for chemical treatment methods to improve interfacial bonding performance between carbon fiber/UHMWPE and core materials<br>5) Certification Testing<br>● Specimen material property testing: DELSEN Testing Laboratories (United States) or SGS North America Inc. (excluding combustion gas and ballistic testing)<br>● Combustion gas testing: VTEC Laboratories (United States) (no domestic certification body available)<br>● Ballistic testing (if required): U.S. standard NIJ Level IIIa; Primary test site: Defense Agency for Technology and Quality (DTaQ), Inje, Gangwon Province, Korea; Secondary test site: Oregon Ballistic Laboratories (United States). |
| **Expected R&R of Korean SME Partner** | 1) Preparation of test and development plans<br>2) Structural design, analysis, and 3D modeling<br>3) Specimen fabrication and testing, and design and fabrication of jigs, fixtures, and molds<br>4) Fabrication of molds for PTM and containers (scaled models)<br>5) Forming and final product assembly<br>6) Acquisition of domestic and international classification and certification (e.g., KR) and marketing to end users (Korea Coast Guard, military, and civilian transport operators)<br>7) Optimization of one-piece forming processes for large PTM structures using the VARTM process<br>8) Fabrication of customized bottom frames and fastening devices compliant with standard specifications for marine containers<br>9) Securing differentiation and originality by using thermal insulation and heat-resistant pads to compensate for the inherent thermal vulnerability of UHMWPE, thereby passing flame and combustion gas tests (Applicable standards: MIL specifications; Test methods: ASTM) Support for optimal formulation and lay-up design of high-performance composite materials (GFRP/CFRP)<br>● Applicable standards: MIL specifications; Test methods: ASTM standards |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Technological Outcomes**<br>1) AI technology development and application<br>2) Securing hybrid structural design and analysis technologies<br>3) Establishing leadership in composite material conversion technologies for ship superstructures and specialized logistics equipment<br>4) Securing modular construction methods for large marine composite structures and localizing core lightweight structural design technologies<br>5) Meeting the requirements of the Republic of Korea Navy (Marine Corps) System-level biodegradable electroceutical platform (TRL 6)<br><br>**Economic Impact**<br>1) Reduction of fuel costs by 15% through ship lightweighting, along with decreased carbon emissions and extended maintenance intervals<br>2) Cost reduction through extended durability, leading to lower operational expenses<br>3) Fuel savings and carbon emission reduction resulting from hull weight reduction, enabling entry into the eco-friendly vessel market through carbon credit acquisition<br>4) Securing commercialization potential, with technology expansion beyond hovercraft to specialized vessel markets such as high-speed passenger ferries and offshore wind farm service vessels (WFSVs) development and application<br><br>**Industrial Impact**<br>1) Securing market leadership in PTMs for military and civilian hovercraft<br>2) Domestic and international entry into the container market for naval vessels and special operations boats of the Navy and Coast Guard<br>3) Expansion into the general marine container market<br>4) Establishing competitiveness in emergency relief and high-speed passenger transportation markets for island and remote regions<br>5) Overseas promotion and sales leveraging research outcomes from Research Institute and official certification reports issued by U.S. testing laboratories |

## 6. References

(1) UHMWPE-Related Patents (Korea)

- KR20160128535A
- KR20100065180A / KR2010
- KR20150034800A

(2) UHMWPE Raw Material Market

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- The global UHMWPE market is expected to grow from USD 2.2 billion in 2024 to USD 3.4 billion by 2029, registering a compound annual growth rate (CAGR) of approximately 9.5% during the forecast period. [Global Information(Global Forecast to 2029)]

(3) Ballistic Composite Materials Market

- GM Insights(2025-2034 report), Coherent Market Insights (2025-2032), Fortune Business Insights
- DAPA and ROK Navy official releases; MOF smart port and island development policies.
- Research and Markets(2025-2033), Fortune Business Insights(2024-2032), Clarkson Research (2024)
- Sources: MOF, The 5th Long-Term Development Plan for the Shipping Industry; KOBC Weekly Market Reports (2025–2026).